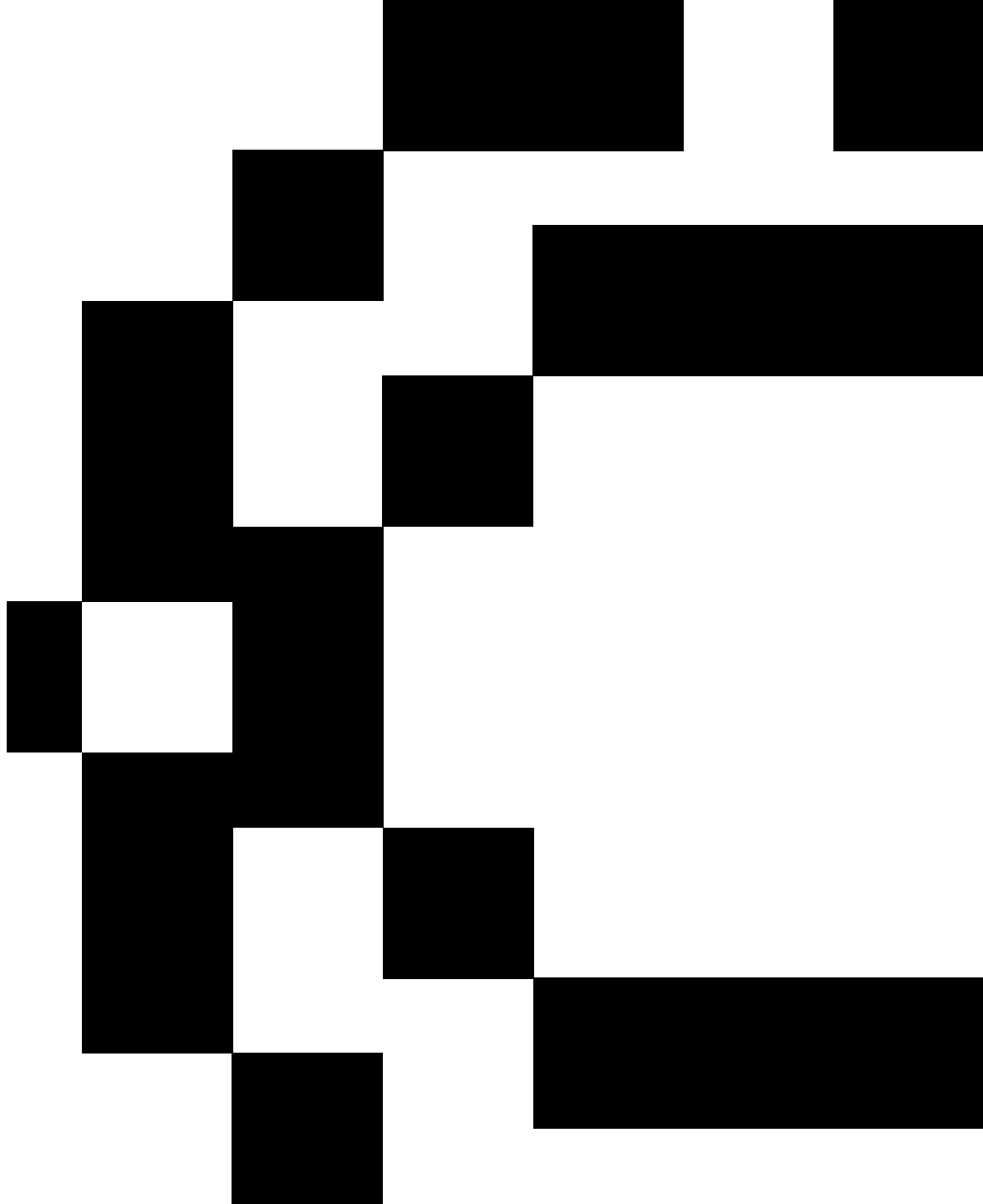
NOVA IMS INFORMATION MANAGEMENT SCHOOL

Computation I / Introduction to Programming

Practicals Week 02 - Data Types and Expressions



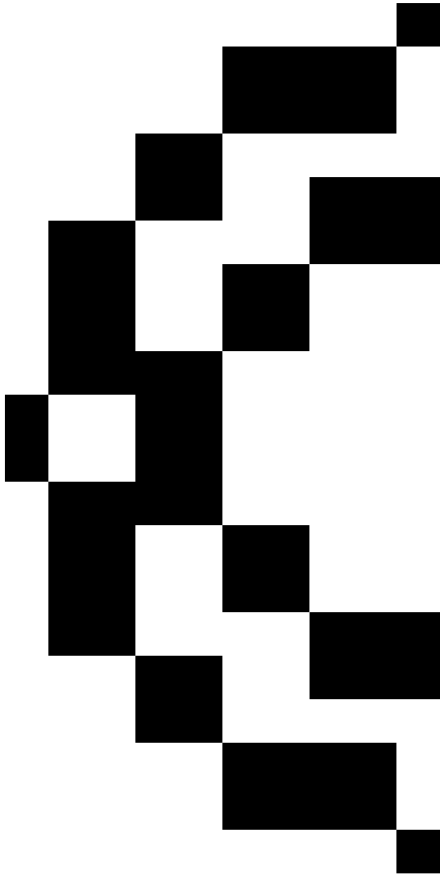
Exercise 1

Given the following variable definitions, write a program that assigns each variable and then evaluates and prints the result of each expression listed below.

a = 2 b = -4.0 c = -3 d = 20.0
e = 3 flag1 = True flag2 = (a == c) f = 20

Expressions to evaluate and print:

a) a + 3 * c - f
b) f / e
c) f % e
d) b * 2 + d / 2 - 2
e) b * (2 + d) / 2 - 2
f) flag1 != flag2
g) 2 > 3
h) not flag1
i) 1 / a
j) 1.0 / a
k) f % 2 + b



Exercise 1 — Solution

```
a = 2
b = -4.0
c = -3
d = 20.0
e = 3
flag1 = True
flag2 = a == c
f = 20

print(a + 3 * c - f)        # a) -27
print(f / e)                 # b) 6.666666666666667
print(f % e)                 # c) 2
print(b * 2 + d / 2 - 2)    # d) 0.0
print(b * (2 + d) / 2 - 2)  # e) -46.0
print(flag1 != flag2)        # f) True  (True != False)
print(2 > 3)                 # g) False
print(not flag1)             # h) False
print(1 / a)                 # i) 0.5
print(1.0 / a)               # j) 0.5  (same as i in Python 3)
print(f % 2 + b)             # k) -4.0  (20 % 2 = 0; 0 + (-4.0) = -4.0)
```

Exercise 2

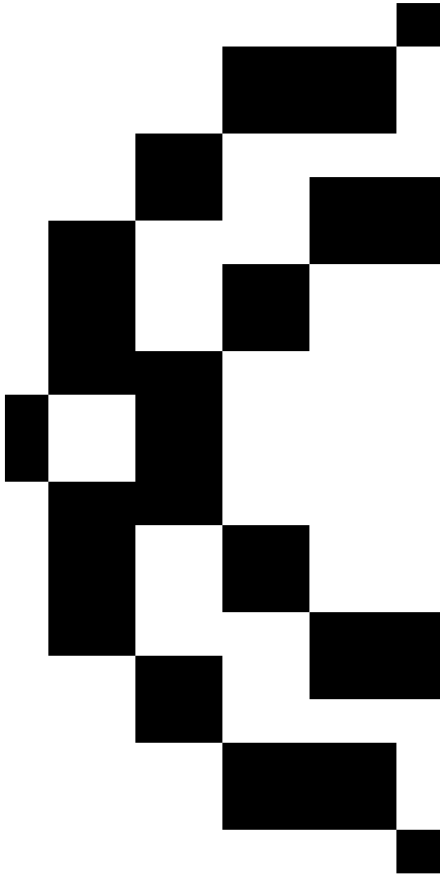
Define two float variables representing the dimensions of a rectangle: `width = 2.0` and `height = 2.5`. Calculate the area and the perimeter, storing each result in its own variable, then print both values.

If you need a refresher, the formulas are as follows:

$$\text{Area} = \text{width} \times \text{height}$$
$$\text{Perimeter} = 2 \times \text{width} + 2 \times \text{height}$$

Example

```
Input:  width = 2.0, height = 2.5
Output: 5.0
        9.0
```



Exercise 2 — Solution

```
width = 2.0
height = 2.5
area = width * height
perim = 2 * width + 2 * height
print(area)   # 5.0
print(perim)  # 9.0
```

Exercise 3

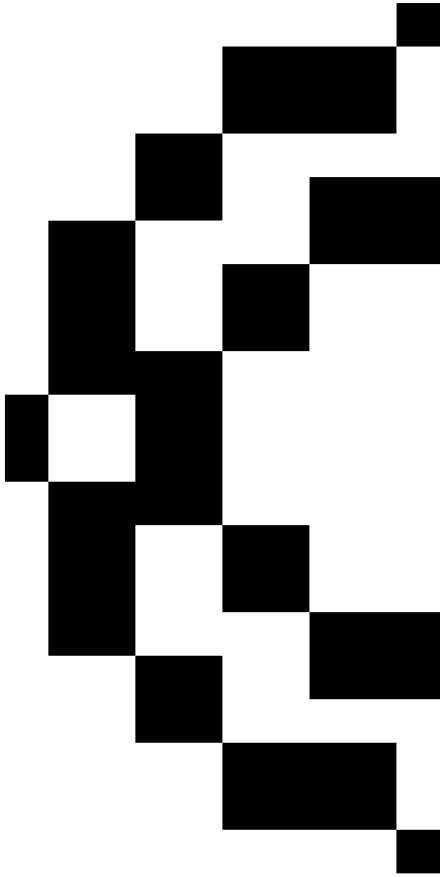
Define two integer variables `c1 = 3` and `c2 = 6`, representing the two shorter sides of a right triangle. Calculate and print the hypotenuse `h` using the Pythagorean theorem:

$$h = \sqrt{(c1^2 + c2^2)}$$

In Python, remember `a ** b` means "a to the power of b" or “a^b”. Consider how to use this for the square root.

Example

- `Input: c1 = 3, c2 = 6`
- `Output: 6.708203932499369`



Exercise 3 — Solution

```
c1 = 3
c2 = 6
sum_of_squares = c1 ** 2 + c2 ** 2
h = sum_of_squares ** (1/2)
print(h)  # 6.708203932499369
```

Exercise 4

Create three integer variables: `var1 = 7`, `var2 = 4`, `var3 = 12`. Print `True` or `False` for each of the following questions:

a) Is `var1` more than three times the sum of `var2` and `var3`?

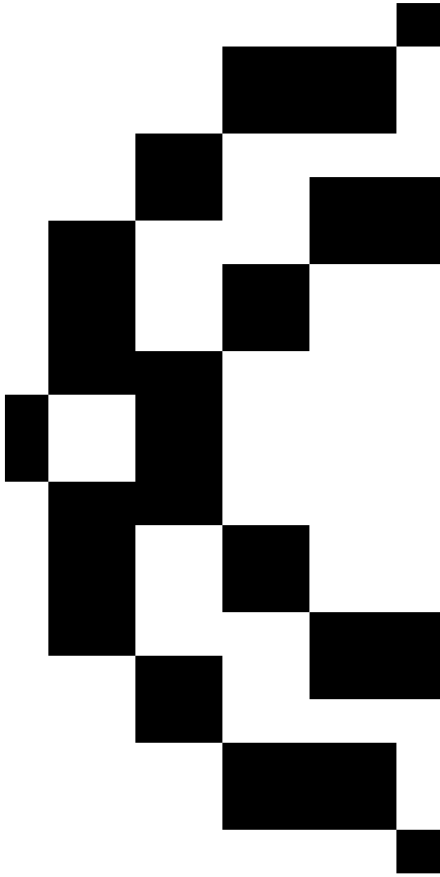
b) Is `var2` even?

c) Is `var3` divisible by `var2`?

d) Is the sum of all three variables greater than 100?

Example

```
Input:  var1 = 7, var2 = 4, var3 = 12
Output: False
        True
        True
        False
```



Exercise 4 — Solution

```
var1 = 7
var2 = 4
var3 = 12

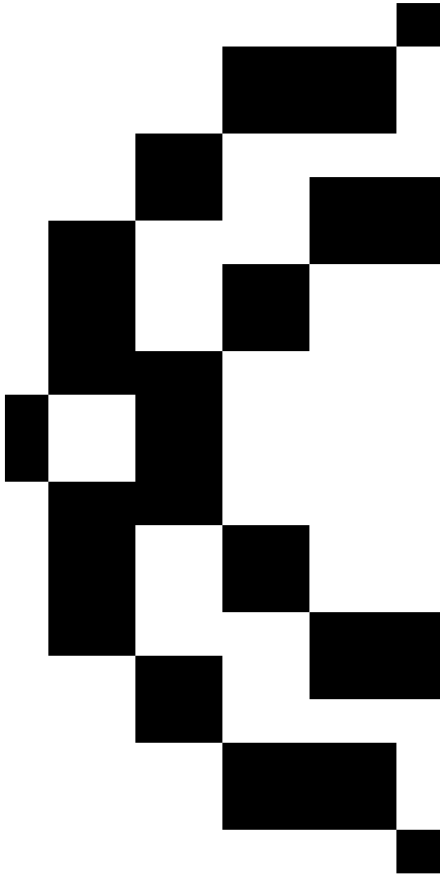
print(var1 > 3 * (var2 + var3))  # a) False  (7 > 48 is False)
print(var2 % 2 == 0)             # b) True   (4 is even)
print(var3 % var2 == 0)          # c) True   (12 is divisible by 4)
print(var1 + var2 + var3 > 100)  # d) False  (23 > 100 is False)
```

Exercise 5

Alexia just arrived from the USA and mentioned it was 40°F. Define a variable `fahr = 40` and convert it to Celsius using the formula below. Store the result in a variable `celc` and print a sentence showing the temperature in both units.
C = (5 / 9) × (F − 32)

Example

- `Input:  fahr = 40`
- `Output: 40 ºF correspond to 4.444444444444445 ºC`



Exercise 5 — Solution

```
fahr = 40
celc = 5 / 9 * (fahr - 32)
print(fahr, "ºF correspond to", celc, "ºC")
# Output: 40 ºF correspond to 4.444444444444445 ºC
```

Exercise 6

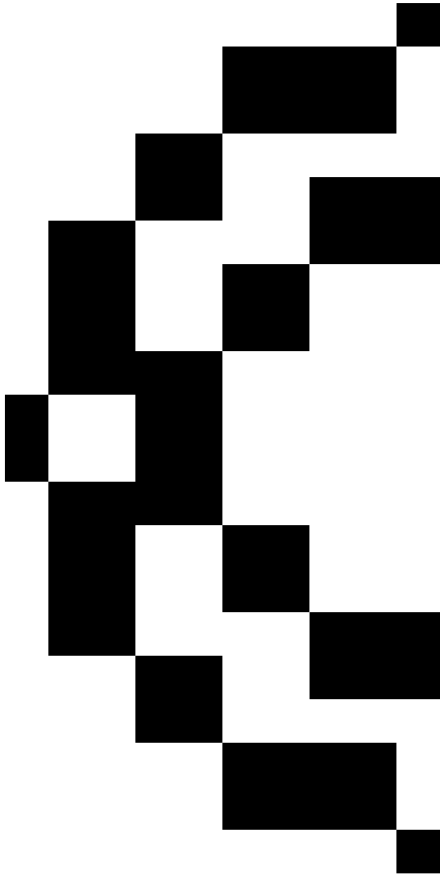
Define an initial amount of money invested, called the principal (in Portuguese, “capital inicial”), of `principal = 1000` euros.

Each month, the bank adds interest to the account equal to `rate / 12` of the current balance, where `rate = 0.05`.

After `years = 3` years (36 months), calculate the final balance (“`saldo`”), storing it in a variable balance, and print it alongside the number of years.

Example

- `Input: principal = 1000, rate = 0.05, years = 3`
- `Output: Balance after 3 years: 1161.4722313334678`



Exercise 6 — Solution

```
principal = 1000
rate = 0.05
years = 3
balance = principal * (1 + rate / 12) ** (12 * years)
print("Balance after", years, "years:", balance)
```

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Thank you

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